

Dense, Sparse, and Hybrid Retrieval for Domain-Specific RAG:

An Evaluation on Synthetic Nginx Technical Documentation

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Abstract

Retrieval-augmented generation (RAG) systems for technical support and system-administration domains must handle both queries that reuse exact technical identifiers (directive names, HTTP status codes) and queries that paraphrase a problem in natural language. We construct a synthetic, original 60-document Nginx administration knowledge base and a 180-question evaluation set balanced across three categories — keyword, semantic (paraphrased), and troubleshooting (multi-concept scenarios) — with programmatically verified category labels. We compare BM25, dense embedding retrieval (all-MiniLM-L6-v2 with FAISS), and a hybrid combination across a full weighting sweep, followed by a cross-encoder reranking stage. Hybrid retrieval ($\alpha=0.75$) achieves the best overall Recall@1 (76.1%), Recall@3 (91.1%), and MRR (0.831), but paired bootstrap and McNemar tests show this improvement over dense-only retrieval (72.2% Recall@1) is not statistically significant at $n=180$. Contrary to our initial hypothesis, dense retrieval already achieves perfect Recall@1 on keyword queries; hybrid retrieval’s gain instead concentrates on semantic and troubleshooting queries. Cross-encoder reranking shows no overall benefit and a category-dependent effect: it improves semantic-query retrieval but substantially degrades troubleshooting-query retrieval, plausibly reflecting a mismatch between the reranker’s web-search training distribution and longer scenario-style technical queries. We report all results, significant and non-significant, and discuss implications for building retrieval pipelines over technical documentation.

Keywords: retrieval-augmented generation, hybrid retrieval, BM25, dense retrieval, reranking, information retrieval evaluation

1. Introduction

Retrieval-augmented generation (RAG) systems answer questions by first retrieving relevant documents from a knowledge base and then conditioning a language model’s response on that retrieved content. Retrieval quality is therefore a primary bottleneck on end-to-end system quality: a language model cannot produce a correct, grounded answer if the retriever fails to surface the document that contains it.

A common concern in technical domains is that dense embedding retrieval, which represents queries and documents as vectors and matches by semantic similarity, may under-weight exact lexical identifiers that matter a great deal in technical text: a specific Nginx directive name (`worker_connections`), an HTTP status code (502), or a specific error string. Classical sparse retrieval methods such as BM25 match on exact term overlap and are, in principle, well suited to exactly this case. This motivates hybrid retrieval, which combines dense and sparse signals, as a natural candidate for technical-documentation RAG.

This paper evaluates that intuition directly and empirically, using a purpose-built, category-labeled evaluation set rather than a single aggregate accuracy number. Our contributions are:

- A 60-document synthetic Nginx administration knowledge base and a 180-question evaluation set, balanced across keyword, semantic, and troubleshooting query categories, with each category label programmatically verified against the target document’s core technical term.
- A full comparison of BM25-only, dense-only, and hybrid retrieval across a complete weighting sweep (α in $\{0.00, 0.25, 0.50, 0.75, 1.00\}$), reported in full rather than at a single cherry-picked operating point.

- Paired statistical significance testing (bootstrap confidence intervals and McNemar’s test) on the retrieval comparisons, which is often omitted in small-scale RAG evaluations.
- A cross-encoder reranking experiment layered on top of the best hybrid configuration, with a category-level breakdown that reveals a reranking effect that is not uniform across query types.

We find that the intuitive story — dense search misses keywords, hybrid search fixes it — is only partly correct, and we report the more nuanced and, we argue, more useful picture that the data actually supports.

2. Related Work

Retrieval-augmented generation was popularized as a way to combine parametric language model knowledge with non-parametric retrieval over an external corpus, and has since been surveyed extensively as the paradigm has been applied across domains and system designs (Gao et al., 2023).

Hybrid retrieval, combining sparse lexical matching with dense semantic matching, has been studied as a way to capture complementary strengths of the two paradigms in open-domain question answering (Arivazhagan et al., 2023). In enterprise and product-support settings, hybrid and dense retrieval baselines have been compared directly for domain-specific question answering, motivating our choice of a similar technical-support-style domain (Eppalapally et al., 2024).

Most closely related to our central finding, recent benchmarking work on heterogeneous technical documents (text-and-table financial filings) found that BM25 outperformed a strong dense retriever (text-embedding-3-large) on most metrics, challenging the general assumption that dense retrieval dominates sparse retrieval, and attributing this to precise domain-specific terminology that lexical matching captures effectively. Our results are consistent with the weaker but related observation that dense retrieval’s advantage over sparse/hybrid methods is not uniform across query types even within a single domain.

On the evaluation methodology side, we follow standard retrieval metrics (Recall@k, MRR) rather than end-to-end generation-quality frameworks such as RAGAS (Es et al., 2023), since this study isolates the retrieval stage of the RAG pipeline rather than evaluating generated answers.

3. Dataset

3.1 Knowledge Base

The knowledge base consists of 60 short documents covering common Nginx system-administration topics: directives (e.g. `worker_connections`, `proxy_pass`, `try_files`), HTTP status codes and their causes (403, 404, 413, 502, 504), and operational topics (reloading configuration, load balancing, rate limiting, SSL/TLS setup, logging). All document text was written originally for this study and does not reproduce official Nginx documentation verbatim, avoiding licensing and copyright concerns while preserving realistic technical vocabulary and phrasing.

3.2 Evaluation Questions

180 questions were constructed, 60 per document on average distributed across three categories (60 questions per category):

- **Keyword:** the question deliberately reuses the target document’s core technical identifier (a directive name or status code), e.g. "How do I fix `worker_connections` limit being hit under load?"
- **Semantic:** a paraphrased description of the same underlying problem that deliberately avoids that specific identifier, sometimes phrased in Hindi-English (Hinglish) code-switched text to further diverge lexically from the source document, e.g. describing a server that "stops accepting new users" rather than naming `worker_connections`.
- **Troubleshooting:** a realistic multi-symptom scenario, intentionally harder and sometimes lexically closer to a neighboring, related document than to the correct target, so that retrieval is not trivially one-to-one.

Each question record includes an id, the question text, its category, an `expected_doc_id` ground-truth label, and a short `reference_answer` paraphrased from (not copied from) the source document, for potential future use in end-to-end answer-quality evaluation.

3.3 Label Verification

Because category labels are only useful if accurate, every keyword question was programmatically checked to contain its target document’s designated core term (e.g. "`worker_connections`", "502", "`try_files`"), and every semantic question was checked to not contain that exact term. This check found and corrected two initial labeling errors before the dataset was

finalized. No such overlap constraint was applied to troubleshooting questions, which are permitted to share incidental vocabulary with neighboring documents by design, reflecting realistic query difficulty.

4. Methods

4.1 Dense Retrieval

Queries and documents are embedded with the all-MiniLM-L6-v2 sentence-transformer model. Retrieval uses exact nearest-neighbor search (FAISS IndexFlatL2) over the 60 document embeddings, so results are deterministic and not subject to approximate-search error.

4.2 Sparse Retrieval (BM25)

BM25 scoring (via the rank_bm25 implementation) is computed over whitespace-tokenized, lowercased document and query text, with no stemming or stop-word removal applied.

4.3 Hybrid Fusion

For a given query, dense similarity scores and BM25 scores are each independently min-max normalized to [0, 1] across the corpus, then linearly combined:

$$\text{hybrid_score} = \alpha \times \text{normalized_dense_similarity} + (1 - \alpha) \times \text{normalized_BM25_score}$$

α is swept over {0.00, 0.25, 0.50, 0.75, 1.00}; $\alpha=1.00$ is mathematically equivalent to dense-only and $\alpha=0.00$ to BM25-only, both included in the sweep as an internal consistency check.

4.4 Cross-Encoder Reranking

As a further stage on top of the best hybrid configuration ($\alpha=0.75$, selected from the sweep), a top-10 candidate shortlist per query is reranked using the ms-marco-MiniLM-L-6-v2 cross-encoder, which scores each (query, document) pair jointly rather than via separately computed embeddings. The reranked top-1/3/5 are then re-evaluated.

4.5 Evaluation Metrics

Recall@k (k in {1, 3, 5}) measures whether the correct document appears anywhere in the top-k retrieved results. Mean Reciprocal Rank (MRR) is computed from a single full top-5 ranked list per query, as $1/\text{rank}$ if the correct document is present in the top 5 and 0 otherwise; MRR is not derived from or constrained to equal Recall@1, since it reflects rank position rather than a fixed cutoff.

4.6 Statistical Significance Testing

For key comparisons, we test the per-query binary Recall@1 outcomes (correct/incorrect) using two paired tests: (i) a paired bootstrap over 10,000 resamples of the 180 questions, reporting the 95% confidence interval of the Recall@1 difference and a one-sided empirical p-value; and (ii) McNemar’s test, which evaluates whether disagreements between two methods (one correct, one incorrect on the same question) are significantly lopsided in one direction.

5. Results

5.1 Overall Retrieval Comparison

Method	Recall@1	Recall@3	Recall@5	MRR
BM25 only	50.6%	69.4%	79.4%	0.610
Dense only (all-MiniLM-L6-v2)	72.2%	85.6%	92.8%	0.801
Hybrid, $\alpha=0.75$ (best)	76.1%	91.1%	93.3%	0.831

Table 1. Overall retrieval performance across all 180 questions.

5.2 Alpha Sweep

The complete sweep is reported below rather than only the best-performing point.

α	Recall@1	Recall@3	Recall@5	MRR
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0.00 (BM25)	50.6%	69.4%	79.4%	0.610
0.25	58.3%	79.4%	87.2%	0.696
0.50	67.8%	87.2%	90.6%	0.774
0.75	76.1%	91.1%	93.3%	0.831
1.00 (Dense)	72.2%	85.6%	92.8%	0.801

Table 2. Full alpha sweep. Recall@1 rises monotonically from alpha=0 to alpha=0.75, then decreases slightly at alpha=1.00, indicating that the optimum is a specific blend rather than "as much dense as possible."

5.3 Category Breakdown

Method (Recall@1)	Keyword	Semantic	Troubleshootin g
BM25 only	85.0%	36.7%	30.0%
Dense only	100.0%	51.7%	65.0%
Hybrid alpha=0.75	100.0%	58.3%	70.0%

Table 3. Recall@1 broken down by question category.

Dense retrieval alone already achieves a ceiling 100% Recall@1 on keyword questions, leaving no room for hybrid retrieval to improve this category. The hybrid method's overall advantage is instead driven entirely by the semantic (+6.6 points over dense) and troubleshooting (+5.0 points) categories.

5.4 Statistical Significance

Comparison	Bootstrap 95% CI on Recall@1 diff	McNemar p
Hybrid (a=0.75) vs. Dense-only	[-1.1%, +8.9%]	0.190
Hybrid+Rerank vs. Hybrid	[-11.1%, +2.2%]	0.243

Table 4. Both confidence intervals include zero and both p-values exceed 0.05: neither comparison is statistically significant at $n=180$.

The hybrid-vs-dense comparison shows a one-sided bootstrap p-value of 0.077 — suggestive of a real effect but not conventionally significant. We report the descriptive advantage of hybrid retrieval throughout this paper while explicitly flagging that it has not been established as statistically significant at this sample size.

5.5 Error Analysis

The best configuration (hybrid, alpha=0.75) failed to place the correct document in the top 3 for 16 of 180 questions (8.9%): 13 semantic, 3 troubleshooting, 0 keyword. Representative failure patterns:

- Near-miss confusions between adjacent error/troubleshooting documents: a paraphrased 504-timeout question failed to retrieve the correct document (504 Gateway Timeout) in the top 5 at all, instead returning several other plausible-sounding technical documents — a genuinely hard case rather than a labeling error.
- A recurring false positive: the "403 Forbidden" document appeared in incorrect top-5 lists across multiple unrelated questions (proxy headers, caching, logging), suggesting its generic troubleshooting phrasing gives it unusually broad lexical/semantic pull — a dataset-level artifact worth noting rather than a retrieval-method failure per se.
- A "root vs. alias" question was missed entirely (absent from the top 5) for a paraphrased 404 scenario, though four of the five retrieved documents were genuinely 404/file-serving-adjacent, indicating the miss reflects close semantic neighborhood confusion rather than an unrelated random retrieval.

5.6 Reranking Experiment

Method	Recall@1	Recall@3	Recall@5	MRR
Dense only	72.2%	85.6%	92.8%	0.801
Hybrid, alpha=0.75	76.1%	91.1%	93.3%	0.831
Hybrid + Cross-Encoder Rerank	71.7%	87.2%	92.2%	0.800

Table 5. Reranking the hybrid top-10 shortlist with a general-purpose cross-encoder did not improve overall performance; all metrics are roughly flat to slightly worse than hybrid alone, and this difference is not statistically significant (Section 5.4).

Method (Recall@1)	Keyword	Semantic	Troubleshootin
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Hybrid alpha=0.75	100.0%	58.3%	70.0%
Hybrid + Rerank	100.0%	63.3%	51.7%

Table 6. Reranking’s effect is category-dependent: it improves semantic-question retrieval (+5.0 points) but substantially degrades troubleshooting-question retrieval (−18.3 points), a pattern invisible in the aggregate result.

6. Discussion

Our initial hypothesis, motivated by prior hybrid-retrieval literature, was that dense retrieval would under-perform on queries containing exact technical identifiers, and that BM25 (via hybrid fusion) would recover this gap. The data does not support that specific mechanism: dense retrieval already achieves perfect Recall@1 on keyword-category questions, so there is no keyword-matching gap for hybrid retrieval to close in this dataset. Instead, hybrid retrieval’s measurable (if not statistically significant) advantage concentrates on semantic and troubleshooting questions — the harder, more paraphrased or multi-concept queries — suggesting that BM25’s contribution here functions less as a keyword-recovery mechanism and more as a complementary, partially redundant signal that adds robustness on queries where dense retrieval alone is already somewhat unreliable.

This is a more specific and, we would argue, more useful finding for practitioners than the original hypothesis: it suggests that for a domain where dense retrieval already handles exact terminology well, the practical value of adding BM25 should be evaluated on the harder, more ambiguous slice of expected queries rather than assumed uniformly.

The reranking result offers a related caution. A general-purpose cross-encoder trained primarily on short, direct web-search-style queries (as in MS MARCO) improved retrieval on paraphrased semantic queries, which resemble that training distribution, but substantially hurt performance on longer, multi-clause troubleshooting queries, which do not. This is consistent with the reranker being distracted by surface-level phrase overlap with incorrect shortlist candidates on longer queries, rather than the reranker being uniformly unhelpful. Aggregate metrics alone (Table 5) would have obscured this and might have led to a premature "reranking doesn’t help here" conclusion; the category breakdown (Table 6) is what makes the effect visible and interpretable.

Finally, the significance testing in Section 5.4 is, in our view, the most methodologically important result in this paper. Reporting only the descriptive numbers (76.1% vs. 72.2%) would suggest a confident conclusion that the data does not, in fact, support at this sample size. We consider explicitly reporting non-significant results, rather than omitting or downplaying them, essential to a scientifically honest small-scale RAG evaluation.

7. Limitations

- The knowledge base is synthetic and single-domain (Nginx administration); results may not generalize to other technical domains, to real (noisier, less consistently structured) documentation, or to multilingual settings.
- At 60 documents and 180 questions (60 per category), this is a small-scale evaluation appropriate for a short paper or workshop submission; the statistical power to detect a true few-percentage-point effect at this sample size is limited, as reflected directly in Section 5.4’s results.
- A single expected_doc_id ground-truth label was assigned per question. A small number of troubleshooting questions could plausibly be answered by more than one document by design (reflecting realistic ambiguity), which makes Recall@k here a slightly conservative estimate of "a useful document was retrieved" versus "the one labeled document was retrieved."
- Results depend on the specific dense embedding model (all-MiniLM-L6-v2) and cross-encoder (ms-marco-MiniLM-L-6-v2) used; larger or domain-adapted models may shift both the magnitude and the category-level pattern of results.
- The "403 Forbidden" document’s recurring false-positive pattern (Section 5.5) suggests at least one dataset-level artifact whose influence on the aggregate numbers has not been separately quantified.
- This study evaluates retrieval only, not end-to-end generated-answer quality; the reference_answer field in the dataset is provided for future work on that question but was not used in the experiments reported here.

8. Conclusion and Future Work

We evaluated dense, sparse, hybrid, and reranked retrieval on a purpose-built, category-labeled technical-documentation dataset. Hybrid retrieval at alpha=0.75 achieved the best descriptive results (76.1% Recall@1, 91.1% Recall@3, 0.831

MRR), but this advantage over dense-only retrieval was not statistically significant at $n=180$. Contrary to our initial hypothesis, the gain was concentrated not in keyword-heavy queries (already handled perfectly by dense retrieval alone) but in semantic and troubleshooting queries. Cross-encoder reranking showed no overall benefit and a category-dependent effect, improving semantic-query retrieval while degrading troubleshooting-query retrieval.

Future work includes scaling the evaluation set and knowledge base to reduce confidence-interval width and better detect true effect sizes; testing additional embedding and reranking models, including domain-adapted or few-shot-prompted rerankers for troubleshooting-style queries; extending the dataset to additional technical domains to test generalization; and evaluating end-to-end generated-answer quality using the `reference_answer` field already included in the dataset.

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Data and Code Availability

The synthetic knowledge base (`docs.jsonl`), the 180-question labeled evaluation set (`qa_final_180.jsonl`), and the Colab notebooks implementing all retrieval methods, the alpha sweep, significance tests, and the reranking experiment are available alongside this paper for reproducibility.